QP CODE: 24900055

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course-MG1DSCCND100

BASIC NUTRITION AND DIETETICS

(2024 ADMISSION ONWARDS)

Duration: 2 HoursMaximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple ChoiceQuestions
Answer all questions
Each question carries 1 mark

1. Ptyalin

2. Glycogen

3.Glycoprotein

4. Vitamin D

5. Meat

6. Clear fluid diet

Part B
Short Answer Type Questions.
Answer any twoquestions.
Each question carries 2 marks.

7.Between 19 and 39 years of age-**0.5 marks**
Weighs 65 kg with a Height of 1.77 m andBMI of 20.75 **-0.5 marks**
Free from disease and physically fit for active work**- 0.5 marks**
On each working day, he is engaged in 8 hours of occupation which usually involves moderate activity-**0.5 marks**

While not at work, he spends 8 hours in bed, 4–6 hours in sitting and moving about, 2 hours in walking, and in active recreation or household duties- **0.5 marks**

8.Regular physical activity can increase muscle mass- **0.75mark**
which in turn increases BMR because muscle tissue requires more energy to maintain than fat- **1.25 mark**

9. Role of vitamin D in small intestine- facilitates the absorption of dietary calcium and phosphorus – ***1 mark***
Role of vitamin D in bone - stimulate the mobilisation of calcium and phosphorus- **0.5 mark**
Role of vitamin D in kidney - reabsorption of calcium – ***0.5 mark***

10.Liquid diet- clear liquid diet, full fluid diet – ***1 mark***

Soft diet- mechanical soft diet, fibre restricted soft diet – ***1 mark***

Part C
Short Essay Type Questions.
Answer any fourquestions.
Each question carries 5 marks.

11.Balanced diet definition-**2 marks**
Dietary sources of carbohydrates in a balanced diet -**2 marks**
Calorie distribution in a balanced diet -**1mark**

12. TER definition-2 marks
Factors influence energy requirement-age, sex, body composition, physical activity and stress-**1 mark**
Distribution of energy needs during basal state- **2 marks**

13.***Role of Vitamin A in vision – 3 marks***

- Retinol is oxidized to retinal which combines with opsin to produce rhodopsin-**1mark**
- When light strikes the retina, the rod cells are bleached, rhodopsin splits to form retinal and opsin and nerve stimulus is transmitted to the visual centre of the brain, where the sensation of vision is created-**2 marks**

Role of vitamin A in growth-2 marks

- Vitamin A, is essential for normal growth and development-**1 mark**
- Vitamin A deficiency leads to loss of appetite, followed by a cessation of growth then rapid weight loss and ultimate death- **1 mark**

14. Release of energy from CHO, protein, fat and alcohol. ***-1 mark***
- Synthesis of protein, fat and pentose sugars for formation of RNA and DNA. - ***1 mark***
- Part of various coenzymes like NAD and NADP. - ***1 mark***
- The coenzymes help in the biological oxidation and in metabolic reaction of CHO. - ***1 mark***

(Examples for enzymatic reactions of niacin as NAD and NADP)- ***1 mark***

15.***Role of vitamin C in collagen synthesis – 5 marks***

- Vitamin C is required by the fibroblasts and the bone forming osteoblasts within bone. – ***1 mark***
- Any deficiency in vitamin C results in defective collagen synthesis, associated with impaired wound healing. - ***1 mark***
- If collagen synthesis is impaired, the matrix formation is defective and it becomes less able to accumulate the calcium and phosphorus required for proper bone mineralisation. - ***1 mark***
- The bone becomes weakened and sometimes distorted.- ***1 mark***
- Bones sometimes become displaced from their joints when the supporting cartilage, which is also mainly collagen, becomes weakened. - ***1 mark***

16.***Answer any 10 points. Each point carries 0.5 marks***

- Plans, develops, controls and evaluates food service systems.
- Develop specification for the procurement of food, equipment and supplies.
- Plans layout designs and determines equipment requirements.
- Recommends and monitors standards for sanitation, safety and security in food service.
- Consults with health care team concerning the nutritional care of patients.
- Provide guidance and evaluation of the job performance of dietetic personnel.
- Assess and evaluate the patient's nutritional status, formulate nutritional care plan, designs meal pattern, individualize according to the patient's food habits, modify according to the therapeutic need and recommends appropriate formula for enteral feeding.
- Has to give individual diet counselling for the patients.
- Prescribes the diet to the patients depending on their disease and plans different menu for different patients suffering from different disease.
- Develop menu pattern and evaluate client acceptance.
- Calculate the nutritive value of the menu and check whether the patient's nutritional needs are met.
- Encourage the patient to consume the modified diet.
- Counselles the patient before discharging regarding the restrictions followed in the diet.
- Plans conduct and evaluate nutrition and health education for a group of patients.

Part D

Essay Type Questions.
Answer any twoquestions.
Each question carries 10 marks.

17.Digesttion, absorption and transport of nutrients-10 marks

- Digestion in mouth-saliva, salivary amylase, lingual lipase-**1 mark**

- Digestion in Stomach-Hydrochloric acid, pepsin, gastric lipase-**2 marks**
- Digestion in small and large intestine-**2 marks**
- Absorption in stomach, small intestine and large intestine-**2 marks**
- Transport-passive diffusion,facilitated diffusion,active transport-**3 marks**

18. BMR-Definition-**1 mark**
    Factors influencing BMR- Genetics,gender,age,weight, body surface area, body fat percentage, diet, body temperature, External temperature, glands, exercise, pregnancy **(any 9 factors-1 mark each)**

19.***Each point carries 1 mark***

1. Helps in vision
2. Helps in growth
3. Helps in Cell differentiation and Gene expression
4. The Regulatory function of Retinoic Acid
5. Needed for immunity
6. Helps in Reproduction
7. Synthesis of glycoproteins
8. Vitamin A and Nerves: needed for myelin sheath
9. Vitamin A and Protein Deficiency
10. Role in RNA and protein metabolism

20.Diet counselling is a combination of nutrition expertise and psychological skills delivered by a trained dietitian. - ***2 marks***
Types- Individual, group, directive, non-directive. – ***4 marks***
Steps- interviewing, counselling, and consulting. - ***4 marks***